GATE

ALL BRANCHES

General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 08

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Recap of Previous Lecture

Topic

Number System

Time and Work

Topics to be Covered

Topic-1 Time and Work

Topic-2 Pipes and Cistern

[MCQ]

#Q. Find the least perfect cube divisible by 2, 3, 4 and 6.

4 & 6

Assignment

A. 216

B. 1728

C. 512

D. 360

[MCQ]

#Q. How many natural numbers up to 100 are divisible by both 2 and 3?

Assignment

$\frac{6}{6 \times 1}$ $\frac{96}{6 \times 16}$

16

A 13

B 14

C 17

D 16

[MCQ]

#Q. Find the largest four digit number that is exactly divisible by 88.

Assignment

$88 = 11 \times 8$

A. 9844

B. 9768

C. 8894

D. 9944

Direct & Inverse:

(+)

(−)

(×)

(÷)

+

−

× E_1

÷ E_2

Time & Work

$\frac{1}{10}$ A = $\frac{10}{1}$ days

$\frac{1}{20}$ B = $\frac{20}{1}$ days

A & B = $\frac{1}{10} + \frac{1}{20}$

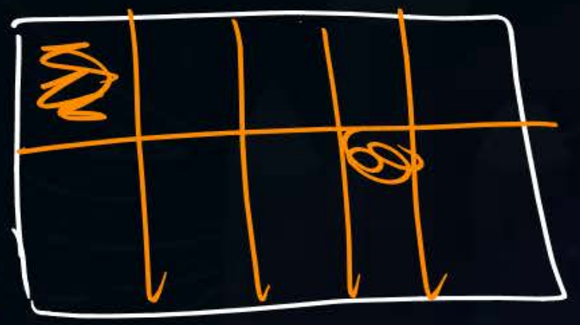
$= \frac{3}{20}$

$\frac{20}{3} = 6.\overline{6}$ days OR $6\frac{2}{3}$ days

TIME — WORK — MEN

INVERSELY

Direct — Direct



Let's Try:

$$\frac{xy}{x+y}$$

Q.1 A ⇒ 20 days

B ⇒ 30 days

$\frac{1}{20}+\frac{1}{30}$

$= \frac{30+20}{20\times30}$

Together = 12 days

Q.2 P ⇒ 25 days | Q ⇒ 40 days

$\frac{20\times30}{30+20}$

Together = $15\frac{5}{13}$ days

Q.3 A & B = 40 days | A = 60 days

B = 120 days

[MCQ]

$A = \frac{1}{10}$ | $B = \frac{1}{20}$

#Q. A can do a work in 10 days whereas B in 20 days.They started working on alternate days, i.e. 1st day 'A', 2nd day 'B', 3rd day 'A'.....so on. Then in how many days the total work would be completed ?

1 2 3 4
A B A B - - -

$\frac{1}{10} + \frac{1}{20} = \boxed{\frac{3}{20}} \rightarrow$ 2 days $\times 6$

$\frac{18}{20} \rightarrow$ 12 days

$\frac{1}{10}$?

13 days

[MCQ]

$A = \frac{1}{10}$ | $B = \frac{1}{20}$

#Q. A can do a work in 10 days whereas B in 20 days.They started working on alternate days, i.e. 1st day 'B', 2nd day 'A', 3rd day 'B'.....so on. Then in how many days the total work would be completed ?

$13\frac{1}{2}$ days or 13.5 days

1	2	3	4
B	A	B	A

$\frac{1}{20} + \frac{1}{10} = \frac{3}{20} \rightarrow 2$ days

$\frac{2}{20} = \frac{1}{10}$?

$\frac{18}{20} \rightarrow 12$ days

13th days

$\frac{1}{10} - \frac{1}{20} = \frac{1}{20}$?

1 work A = 10

$\frac{1}{20}$ " A = $10 \times \frac{1}{20}$

$= \frac{1}{2}$

[MCQ]

#Q. A can do a work in 25 days whereas B in 40 days. They started working on alternate days, i.e. 1st day 'A', 2nd day 'B', 3rd day 'A'.....so on. Then in how many days the work would be completed?

A. $15\,^{5}/_{8}$ days

B. $30\,^{5}/_{8}$ days

C. 31

D. 16

[MCQ]

$A = \frac{1}{20}$ | $B = \frac{1}{30}$ | $C = \frac{1}{60}$

#Q. A can do a work in 20 days whereas B in 30 days and C in 60 days. They started the work together. But A left after two days and B left three days before the work got completed. Then the total work was done in how many days?

20 days

$A\&B\&C = \frac{1}{20} + \frac{1}{30} + \frac{1}{60}$

$= \frac{6}{60} = \frac{1}{10}$

$B\&C = \frac{1}{30} + \frac{1}{60} = \frac{3}{60} = \frac{1}{20}$

| work B&C = 20

$\frac{3}{4}$ " B&C = $20 \times \frac{3}{4} = 15$

2 days	? $\frac{3}{4}$	3 days
A&B&C	B&C	C
$\frac{2}{10}$	15 days	$\frac{3}{60} = \frac{1}{20}$

$\frac{2}{10} + \frac{1}{20} = \frac{5}{20} = \frac{1}{4}$

[MCQ]

#Q. A can do a work in 20 days whereas B in 30 days and C in 60 days. They started the work together. But A left after two days and B left three days before the work got completed. Then the total work was done in how many days?

'x' days

$$\frac{x}{60}+\frac{x-3}{30}+\frac{2}{20}=1$$

$$\Rightarrow x+2x-6+6=60$$

$$3x=60$$

20 days

[MCQ]

$A = \frac{1}{10}$ $B = \frac{1}{20}$ $C = \frac{1}{60}$

#Q. A can do a work in 10 days whereas B in 20 days and C in 60 days. They started the work together. But A left at the end of 3rd day and B left at the end of 5th day. Then the remaining work was done by C in how many days?

(A) 27

(B) 35

(C) 31

(D) 22

$$\frac{x}{60} + \frac{5}{20} + \frac{3}{10} = 1$$

$$x + 15 + 18 = 60$$

$$x = 27$$

3 days	2 days	?
A&B&C	B&C	C

'x' days

$27 - 5 = 22$ days

[MCQ]

#Q. If 9 MONKEYS EAT 9 BANANAS IN 9 MINUTES, THEN HOW MANY MONKEYS WILL EAT 45 BANANAS IN 45 MINUTES?

Monkey	Bananas	Minutes
9	9	9
?	45	45

$$9 \times \frac{45}{9} \times \frac{9}{45}$$

Chain Rule:

$\uparrow 1 \times \frac{3}{2}$

$\downarrow 1 \times \frac{2}{3}$

%	No
28% →	61
43% →	?

$61 \times \frac{43}{28}$

28% of $x = 61$

$x = 61 \times \frac{100}{28}$

43% of $x = \frac{43}{100} \times 61 \times \frac{100}{28}$

[MCQ]

#Q. 12 men can do a work in 15 days working 8 hours a day. In how many days can 9 men do the same work, working 10 hours a day?

Men	Days	Hrs/day
12	15	8
9	?	10

$$15 \times \frac{8}{10} \times \frac{12}{9} = 16 \text{ days}$$

[MCQ]

#Q. A hostel with 600 students had sufficient food for 210 days. After 30 days, 240 students left the hostel. Now the remaining food will last for how many days?

330 → T.F.

Extra ↓

120

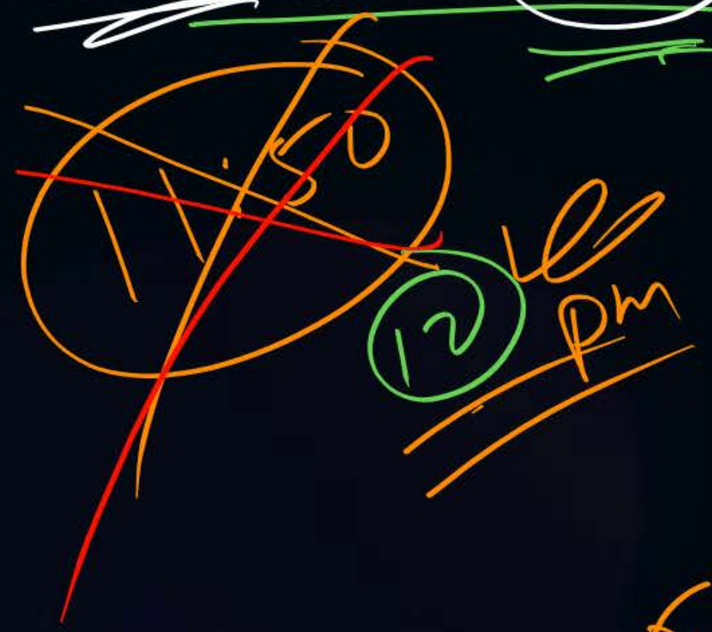
Students	Days
600	210
600	180
360	?

$180 \times \frac{600}{360}$

300 days

[MCQ]

#Q. A Clock which gains 10 minutes in every one hour was set correct at 6 am. If that clock represents 1 pm the same day, what must be the correct time?



Time

C.W	W.W
60	70
?	7 hrs

$60 \times \frac{7}{70}$

$= 6$ hrs

[MCQ]

#Q. A Clock which gains 10 minutes in every one hour was set correct at 6 am. If the correct time is 12 pm the same day, what would be the time shown by that clock?

1 PM

[MCQ]

#Q. A contractor under takes to make a road in 40 days and employs 25 men. After 24 days, he finds that only one-third of the road is made. How many extra men should he employ so that he is able to complete the work 4 days earlier?

A. 75

B. 100

C. 50

D. None of these

$36 - 24 = 12$

Men	Days	Work
25	24	$\frac{1}{3}$
?	12	$\frac{2}{3}$

$25 \times \frac{24}{12} \times \frac{2}{3} \times \frac{3}{1}$

100 Men

[MCQ]

#Q. A can do a work in 60 days whereas B in 20 days. If they together took a contract of ₹32000, then what would be the share of A?

A : B ⇒ 6 : 2

3 : 1 (days)

Efficiency

A : B = 1 : 3

$\frac{1}{4} \times 32,000$

8000

A ₹ 8000

B ₹ 6000

C ₹ 1800

D ₹ 2040

2 6000

A : B = 5 : 7 (days)

= 7 : 5 (Efficiency)

A B

P : Q = 9 : 1 (Efficiency) 10 days 20 days

= 1 : 9 (days)

A : B = $\frac{1}{10}$: $\frac{1}{20}$

A : B = 10 : 20

= 1 : 2

A : B : C = 1 : 2 : 3 (days)

A : B : C = 6 : 3 : 2 (Efficiency) 2 : 1

$$\frac{1}{x} \qquad \frac{1}{y} \qquad \frac{1}{z}$$

Pipes & Cistern

$A : B : C : D = 1 : 2 : 3 : 4$ (days)

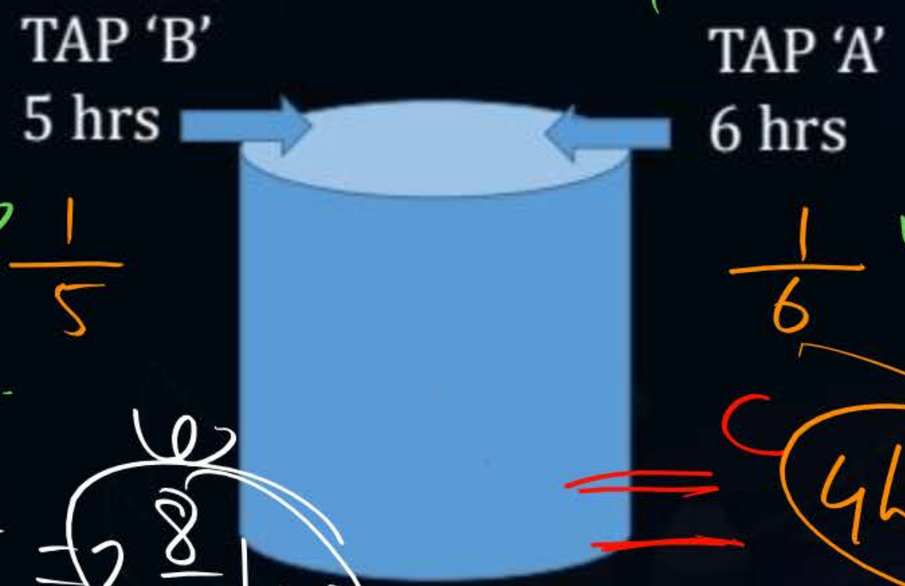
$= 24 : 12 : 8 : 6$

$= 12 : 6 : 4 : 3$

Pipes & Cistern

Empty (−)

filling (+)



$\frac{1}{5}$

$\frac{1}{6}$

4hrs

$-\frac{1}{4}$

A & B → $\frac{1}{6}+\frac{1}{5}$

$= \frac{11}{30}$ → $\frac{30}{11} = 2\frac{8}{11}$ hrs

$\frac{1}{5}+\frac{1}{6}-\frac{1}{4} = \frac{7}{60}$ → $\frac{60}{7} = 8\frac{4}{7}$ hr

[MCQ]

#Q. Two pipes A and B can fill a tank in 8 minutes and 12 minutes respectively. Both the pipes are opened together and after 3 minutes, pipe A is turned off. What is the total time required to fill the tank?

Assignment

A. 3 minutes

B. 7.5 minutes

C. 10 minutes

D. 4.5 minutes

[MCQ]

#Q. Pipes A and B can fill a tank in 4 hours and 8 hours respectively. Pipe C can empty it in 16 hours. If all the three pipes are opened together, then how long will it take to fill the tank?

Assignment

A. 4.5 hours

B. 2 hours

C. 6 hours

D. 3.2 hours

[MCQ]

#Q. A tank can be filled by 20 buckets each of capacity 13.5 litres. If the capacity of each bucket be 9 litres, how many buckets will fill the same tank?

Assignment

A. 30

B. 25

C. 20

D. 15

2 mins Summary

Topic Time and Work

Pipes and Cistern

THANK - YOU